# CS781 Capstone Project: In-House Legal Intelligence System

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**Abstract**

This capstone studies an in-house legal NLP workflow across seven task phases: legal language modelling, named entity recognition, rhetorical-role prediction, court-judgment prediction, statute identification, abstractive summarization, and prior-case retrieval. A LEGAL_40K tokenizer and multiple Phase-1 architecture families were developed and evaluated. Codabench submissions were made for Phases 1, 4, 5, 6, and 7; Phases 2 and 3 were not submitted. The recorded leaderboard ranks were 128, 124, 124, 120, and 89 respectively. The corresponding primary scores were 0.8142, 0.5935, 0.2818, 0.8157, and 0.3420. This report distinguishes these official leaderboard values from local validation metrics and documents the incomplete phases and submission-count shortfall.

## 1 Competition Result

**Codabench username:** m_240671 **Roll number:** 240671

Table 1: Official Codabench leaderboard snapshot recorded on 9 November 2026.

| Phase | Task | Primary metric | Score | Rank | Finished IDs |
|---|---|---|---|---|---|
| 1 | Legal LM Pretraining | BERTScore F1 | 0.8142 | 128 | 5 |
| 2 | Legal NER | Not submitted | – | – | 0 |
| 3 | Rhetorical role prediction | Not submitted | – | – | 0 |
| 4 | Court judgment prediction | Macro-F1 | 0.5935 | 124 | 3 |
| 5 | Legal statute identification | Macro-F1 | 0.2818 | 124 | 6 |
| 6 | Abstractive summarization | BERTScore F1 | 0.8157 | 120 | 7 |
| 7 | Prior Case Retrieval | Micro-F1@7 | 0.3420 | 89 | 7 |

The Phase-1 leaderboard row also reports ROUGE-L 0.1847 and BLEU-4 0.0275. The leaderboard screenshots identify the best entries as submission IDs 964649 (Phase 1, 6 October 2026), 1009219 (Phase 4, 26 October 2026), 1016613 (Phase 5, 28 October 2026), 1030833 (Phase 6, 3 November 2026), and 1040718 (Phase 7, 7 November 2026). Phase 2 and Phase 3 were not submitted. The instructor announcement states that phases without a leaderboard submission are treated as non-participation.

The assignment requires at least four distinct submissions in Phase 1 and six in each of Phases 2–7. The uploaded My Submissions screenshots show five finished Phase-1 entries (one additional entry failed), three finished Phase-4 entries (one additional entry was cancelled), six finished Phase-5 entries, seven finished Phase-6 entries, and seven finished Phase-7 entries. Thus Phase 1 meets the numerical minimum if the five finished entries are distinct; Phase 4 is below the six-entry minimum; Phases 5–7 meet the numerical minimum if their entries are distinct. The screenshots show 28 finished submission IDs in total, but do not independently prove that all prediction contents were distinct. The distinct-submission rule must be checked against the actual uploaded prediction files if any repeated dumps were used.

## 2 Introduction and Problem Description

The capstone assignment is the development of an in-house legal intelligence workflow for Indian legal text. Its seven benchmark phases cover both language generation and legal document understanding. The phases use different prediction contracts and primary metrics: Phase 1 generates a continuation for a legal prompt and is evaluated by BERTScore F1; Phase 2 predicts character-level legal entity spans and uses strict span Macro-F1; Phase 3 assigns rhetorical-role labels to legal sentences and uses Macro-F1; Phase 4 predicts a court-judgment outcome and uses Macro-F1; Phase 5 predicts applicable statute IDs as a multi-label task and uses Macro-F1; Phase 6 generates abstractive legal summaries and uses BERTScore F1; and Phase 7 ranks prior cases and uses Micro-F1@7.

The Phase-1 foundation-model constraint required a compact model trained from random initialization within a 50M–300M parameter range, using the official legal corpus and no externally pretrained language model. The competition also imposed exact output schemas, official test-ID matching, and per-phase submission minimums. The student submitted Codabench predictions for Phases 1, 4, 5, 6, and 7. Phases 2 and 3 were not submitted and are reported as non-participation rather than as completed experiments.

## 3 Data Analysis and Preprocessing

### 3.1 Sources and official splits

The Phase-1 corpus comprises four released legal-text sources: Acts, High Courts, Supreme Court, and Tribunals. The retained token-cache inventory records 152,342,527 tokens for Acts, 1,961,464,909 for High Courts, 258,064,342 for Supreme Court, and 1,502,494,787 for Tribunals, totalling 3,874,366,565 cached trainable tokens. These are recovered cache counts; they should not be interpreted as the total corpus size or as the number consumed by every model run.

Phase 4 uses the official Court Judgment Prediction data; Phase 5 uses case documents paired with one or more statute IDs; Phase 6 pairs legal documents with reference summaries for training; and Phase 7 supplies query cases and split-specific candidate pools. The Phase-6 notebook identifies 7,030 training rows and 100 test rows. The Phase-7 preflight records 827 Train queries with 4,320 candidates, 118 Dev queries with 1,023 candidates, and 237 Test queries with 1,727 candidates. The retained local Phase-7 run reports that 817 Train queries had at least one relevant candidate and all 118 Dev queries were labelled.

Table 2: Recovered Phase-1 token-cache inventory.

| Source | Tokens |
|---|---|
| Acts | 152,342,527 |
| High Courts | 1,961,464,909 |
| Supreme Court | 258,064,342 |
| Tribunals | 1,502,494,787 |
| Total | 3,874,366,565 |

Table 3: Phase-7 split inventory from the executed data preflight.

| Split | Queries | Candidates | Label status |
|---|---|---|---|
| Train | 827 | 4,320 | 817 with positive overlap |
| Dev | 118 | 1,023 | 118 labelled |
| Test | 237 | 1,727 | labels private |

### 3.2 Preprocessing

The project used the 40,000-entry LEGAL_40K tokenizer. The retained tokenizer SHA256 is recorded as 3e38dd5c561d84ad8fbadf792bbca490f8c70346d922df7afbef37c5a942b71e (40,000-entry vocabulary). The Phase-1 notebooks construct token caches from the four official sources and use resumable shards. The Architecture-A training configuration uses a 2,048-token context. For the Phase-1 submission, the generation pipeline validates the official test IDs and the exact 512-whitespace-word prompt contract, and checks that each continuation contains exactly 128 whitespace-separated words.

For Phase 4 and Phase 5, the submitted outputs are keyed by official case IDs. Phase 6 preserves document IDs through summarization generation. Phase 7 uses the official candidate pools and validates the top-seven candidate-ID output. No Phase-2 or Phase-3 preprocessing pipeline was submitted; this report does not claim that character-offset alignment or sentence-level prediction was completed.

## 4 Approach and Model Evolution

### 4.1 Phase-1 foundation model

The retained Phase-1 work explored four compact legal language-model families using LEGAL_40K. Architecture A (LegalDecoderA) is a decoder-only causal Transformer with a recorded parameter count of 243,897,344. Architecture B (LegalHybrid150) has 149,902,080 parameters and uses RMSNorm, RoPE, grouped-query attention, hybrid local/global attention, SwiGLU, residual connections, and tied token embeddings. Its recorded run processed 499,974,144 tokens in 7,629 optimizer steps and reported validation loss 2.8204395548 and perplexity 16.7842266375.

Architecture C (DMLC-150) has 149,061,510 parameters and uses an encoder–decoder design with token-level Evidence Memory and 16 learned Case Memory slots. The recorded run reached step 8,120 and 332,595,200 tokens, or 66.52% of the stated 500M-token target; it did not complete that target. Architecture D (LegalPrefixFormer-145) has 144,786,688 parameters and uses a PrefixLM attention pattern with a 1,024-token prefix and a 256-token supervised target. Its retained technical report shows early training progress but does not evidence an official D-specific BERTScore result.

Table 4: Phase-1 architecture facts documented in the retained technical reports.

| Model | Parameters | Documented design |
|---|---|---|
| A / LegalDecoderA | 243.90M | Decoder-only causal Transformer; 2,048-token context in the retained run |
| B / LegalHybrid150 | 149.90M | Hybrid local/global attention, GQA, RoPE, RMSNorm, SwiGLU |
| C / DMLC-150 | 149.06M | Encoder–decoder with Evidence Memory and 16 Case Memory slots |
| D / LegalPrefixFormer-145 | 144.79M | PrefixLM; 1,024-token prefix and 256-token target |

### 4.2 Transfer protocol (Phases 2–7)

The retained downstream notebooks use Phase-1 checkpoints as initial model assets where available. The exact checkpoint-to-submission mapping must be read from each experiment manifest before a reproducibility claim is made. Phases 2 and 3 were not submitted; no transfer experiment or task head is claimed for those phases.

### 4.3 Phase 2 - Legal NER

The task specification requires character-level legal entity spans with one of 12 entity labels and strict span-boundary matching. No Phase-2 prediction was submitted to Codabench. The phase is therefore

recorded as non-participation and has no score or rank.

### 4.4 Phase 3 - Rhetorical role prediction

The task specification requires rhetorical-role labels for legal sentences from Corporate Law and Information Technology material. No Phase-3 prediction was submitted to Codabench. The phase is therefore recorded as non-participation and has no score or rank.

### 4.5 Phase 4 - Court judgment prediction

The retained CJPE notebooks include an emergency multi-checkpoint workflow and a later full-data follow-up. The local experiment log records an Architecture-A bottleneck-adapter development Macro-F1 of 0.550745. The official Codabench leaderboard score for the selected entry is Macro-F1 0.5935, rank 124, submission ID 1009219. The leaderboard row also displays two additional values, 0.6503 and 0.4514; the supplied crop does not show their column labels, so they are not assigned names here.

### 4.6 Phase 5 - Legal statute identification

The retained LSI notebook evaluates parameter-efficient and layerwise methods using Phase-1 checkpoints. The local Dev record includes an Architecture-C partial-layerwise Macro-F1 of 0.24171. The official Codabench score is Macro-F1 0.2818, Micro-F1 0.3618, and subset accuracy 0.0091, rank 124, submission ID 1016613.

### 4.7 Phase 6 - Abstractive summarization

The retained summarization notebook includes adapter, LoRA, prefix-tuning, and knowledge-distillation experiments. The local E7 scorer reports BERTScore F1 0.782209, ROUGE-1 0.167088, and ROUGE-L 0.133178. E8 reports BERTScore F1 0.815720, ROUGE-1 0.232672, and ROUGE-L 0.140314 over 100 predictions. The selected Codabench result is BERTScore F1 0.8157, ROUGE-1 0.2327, and ROUGE-L 0.1403, rank 120, submission ID 1030833.

### 4.8 Phase 7 - Prior Case Retrieval (PCR)

The retained PCR notebooks document both a multi-experiment pipeline and a long Architecture-A retrieval run. The local Architecture-A Dev metrics are Micro-F1@7 0.341301, Precision@7 0.371671, and Recall@7 0.315519. The selected Codabench result is Micro-F1@7 0.3420, Micro-Precision@7 0.3490, and Micro-Recall@7 0.3353, rank 89, submission ID 1040718.

## 5 Experiments and Implementation Details

### 5.1 Compute and training protocol

The submitted work was developed in Kaggle notebook environments using Tesla T4 GPUs. Retained notebook outputs record two-T4 configurations for some Phase-1 runs. Phase-1 training uses mixed precision, gradient accumulation, periodic validation, and resumable checkpoints. Architecture B processed 499,974,144 tokens over 7,629 steps. Architecture C processed 332,595,200 tokens before the recorded run ended. Architecture D's retained early run processed 12,288,000 tokens by step 300. The phase-specific notebooks contain the experiment configurations; this summary does not infer missing settings from architecture names.

### 5.2 Local evaluation

Local development evaluation and Codabench leaderboard evaluation are reported separately. Local values retained in the notebooks include Phase-4 Dev Macro-F1 0.550745, Phase-5 Dev Macro-F1 0.24171, Phase-7 Dev Micro-F1@7 0.341301, and the Phase-6 E7/E8 scoring values above. The official leaderboard values are listed in Section 1. The two types of metrics should not be mixed because the data split and evaluation context differ.

### 5.3 Ablations and controlled comparisons

The experiment records include full/partial fine-tuning, bottleneck adapters, LoRA, prefix tuning, and knowledge distillation variants. The retained evidence does not provide a complete controlled one-factor ablation table for every participating phase. The student must identify which comparisons were controlled and explain their interpretation using the original experiment logs.

## 6 Results

Table 5: Official leaderboard and local development results. Local values are explicitly distinguished from Codabench values.

| Phase | Evaluation / variant | Result |
|---|---|---|
| 1 | Codabench BERTScore F1 / rank | 0.8142 / 128 |
| 1 | Architecture B validation loss / perplexity | 2.82044 / 16.78423 |
| 4 | Codabench Macro-F1 / rank | 0.5935 / 124 |
| 4 | Local Dev Macro-F1, A bottleneck adapters | 0.550745 |
| 5 | Codabench Macro-F1 / rank | 0.2818 / 124 |
| 5 | Local Dev Macro-F1, C partial-layerwise | 0.24171 |
| 6 | Codabench BERTScore F1 / rank | 0.8157 / 120 |
| 6 | Local E8 BERTScore / ROUGE-1 / ROUGE-L | 0.815720 / 0.232672 / 0.140314 |
| 7 | Codabench Micro-F1@7 / rank | 0.3420 / 89 |
| 7 | Local Dev Micro-F1@7 | 0.341301 |

The submitted phases show that performance depends on the task-specific objective and evaluation metric rather than on a single model configuration. The strongest recorded Codabench entries were Phase 1 (0.8142), Phase 4 (0.5935), Phase 5 (0.2818), Phase 6 (0.8157), and Phase 7 (0.3420). These values are not directly comparable across phases because each phase uses a different task and metric. The local development values are useful for model selection, while Codabench values are the official external evaluation. Differences between them can arise from data distribution, preprocessing, decoding, checkpoint selection, and the distinction between local development and hidden test data. The retained records support the selected workflows listed in the results table, but do not provide enough prediction-level evidence to attribute every score difference to one specific modelling decision.

## 7 Error Analysis and Discussion

The retained engineering logs record several implementation issues. Phase 1 included an earlier test-ID mismatch and subsequent corrections to use the official test split; Phase 4 included checkpoint compatibility and data-discovery debugging; Phase 5 recorded dtype-related experiment failures; Phase 6 documented an unavailable Architecture-D variant because a compatible checkpoint was not present; and Phase 7 had an early run interrupted by a missing metric helper before the later resumable Architecture-A run. These are observed engineering events, not a complete prediction-level error analysis.

### 7.1 Phase-wise failure modes

The supplied archive does not contain a complete prediction-level error file with traceable examples and counts for every submitted phase. I therefore do not report fabricated examples or error frequencies. The engineering records do document implementation issues: a test-ID mismatch in the early Phase-1 workflow, checkpoint and data-discovery issues in Phase 4, dtype-related failures in Phase 5, an unavailable compatible Architecture-D checkpoint in Phase 6, and a missing metric helper during an early Phase-7 run. These are workflow failures, not evidence of model prediction errors.

### 7.2 What the metrics do not measure

The benchmark metrics measure agreement with reference annotations but do not fully establish legal correctness. For NER and rhetorical-role tasks, a high token-level score can still hide boundary mistakes or an incorrect legal interpretation. For summarization, overlap and embedding-based metrics may reward lexical similarity without proving that every proposition is supported by the source judgment. For prior-case retrieval, top-seven ranking metrics do not by themselves establish that a retrieved precedent is legally controlling, factually analogous, or still good law. Human legal review would therefore be required before using these systems in a real legal workflow.

### 7.3 Limitations and what you would do next

The retained package does not include verified competition-model implementations for Phases 2 and 3, so those phases cannot be represented as completed submissions. The project records also indicate that the Phase-4 submission count was below the stated minimum; this should be checked against the official competition account and rules. The supplied notebooks contain no saved execution outputs, so the code package alone cannot reproduce the recorded leaderboard results without the required datasets, checkpoints, credentials, and runtime environment. The next steps are to recover the missing phase implementations if they exist, verify every leaderboard entry and submission-count requirement, preserve prediction-level validation outputs, and rerun the selected notebooks from clean environments.

### 7.4 Surprising findings

One notable result in the retained records is the Phase-6 E8 BERTScore F1 of 0.815720, compared with the recorded E7 value of 0.782209. The Phase-6 notebook describes E8 as a decoding-focused continuation using the E7 adapter rather than a new gradient-training run. This suggests that decoding choices can materially affect measured summarization quality, although the supplied files do not include enough paired example-level analysis to determine which output changes produced the gain.

## 8 Conclusion

This project developed and evaluated legal-language workflows across several stages of the CS781 benchmark. The retained records document submissions for Phases 1, 4, 5, 6, and 7, with the strongest recorded results reported separately for each phase because their metrics are not directly comparable. The work also shows the practical importance of reliable data discovery, checkpoint management, reproducible execution, and task-specific evaluation. Phases 2 and 3 do not have verified competition-model implementations in the supplied archive, and the Phase-4 submission-count requirement remains a point that must be checked against the official competition record. The notebooks are provided as the selected implementation artifacts, but they require the original datasets, checkpoints, and runtime setup to execute. The most important remaining step is to verify the submission record and ensure that every claim in this report matches the official Codabench results and the student's own project records. **Public code repository:**

https://github.com/RK-cyber5/CS781-Capstone.

## LLM Usage Declaration

**Q1.** Which LLM(s) were used?
Answer: ChatGPT.

**Q2.** Mark each row truthfully and state the purpose. The rows below are a disclosure prompt, not a declaration that the uses were permitted.

| Part | Used? | Purpose / confirmation |
|---|---|---|
| Understanding concepts / papers | Yes | Explanations of model and NLP concepts |
| Ideation / brainstorming | Yes | Discussion of project and experiment ideas |
| Debugging ancillary snippets | Yes | Notebook and runtime debugging |
| Plotting / visualization code | To confirm | Yes; plotting and visualization utilities appear in the retained notebooks. |
| Grammar / writing style | Yes | Report wording/formatting support |
| Report structure / section wording | Yes | Report outline and section organization |
| Designing model architecture | No | Designed independently by the student |
| Hyperparameter search / tuning | No | Search strategy, parameter choices, and tuning performed independently by the student |
| Analysis of results / conclusions | Yes | AI assistance was requested; this is outside permitted examples |
| Writing core training / inference code | Yes | AI assistance generated/revised substantial notebook code |
| Other | Yes | Recovery, notebook assembly, documentation, and packaging |

**Q3.** Did you refer to sources other than lecture slides and an LLM?
Answer: Yes. The official CS-781 Capstone Document, official dataset/task instructions, project starter materials, architecture technical reports, and the cited model/evaluation papers were used. The implementation also uses the libraries and model interfaces imported in the notebooks. The student should ensure that all externally copied code, pretrained model sources, and documentation are cited in the repository README and bibliography.

**Q4.** Apart from permitted LLM use, what was your contribution?
I independently designed the model architectures and made the hyperparameter-search and tuning decisions. I organized and maintained separate workflows for the benchmark phases represented in this submission. I worked with the official task data and model checkpoints required by the individual notebooks. I reviewed the architecture alternatives and selected the Phase-1 Architecture-A workflow. I also maintained phase-specific workflows for CJPE, statute identification, summarization, and prior-case retrieval. The project records show that I addressed practical issues involving dataset discovery, test identifiers, checkpoint compatibility, data types, and resumable execution. I collected the relevant notebooks and supporting reports into a phase-organized submission package. I used ChatGPT for explanations, debugging support, assistance with some code, report organization, and drafting; these uses are disclosed in this declaration. ChatGPT did not design my model architectures or perform my hyperparameter search or tuning. I am responsible for checking the implementation, verifying the reported results against Codabench, and ensuring that the final submission complies with the course rules.

## References


[1] A. Vaswani et al., "Attention Is All You Need," NeurIPS, 2017.

[2] N. Houlsby et al., "Parameter-Efficient Transfer Learning for NLP," ICML, 2019.

[3] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," ICLR, 2022.

[4] X. L. Li and P. Liang, "Prefix-Tuning: Optimizing Continuous Prompts for Generation," ACL-IJCNLP, 2021.

[5] G. Hinton, O. Vinyals, and J. Dean, "Distilling the Knowledge in a Neural Network," NeurIPS Workshop, 2015.

[6] T. Zhang et al., "BERTScore: Evaluating Text Generation with BERT," ICLR, 2020.

[7] C.-Y. Lin, "ROUGE: A Package for Automatic Evaluation of Summaries," 2004.

[8] K. Papineni et al., "BLEU: a Method for Automatic Evaluation of Machine Translation," ACL, 2002.

[9] Exploration Lab, IIT Kanpur, "CS-781 Capstone: In-House Legal Intelligence System," course project document, 2026.